

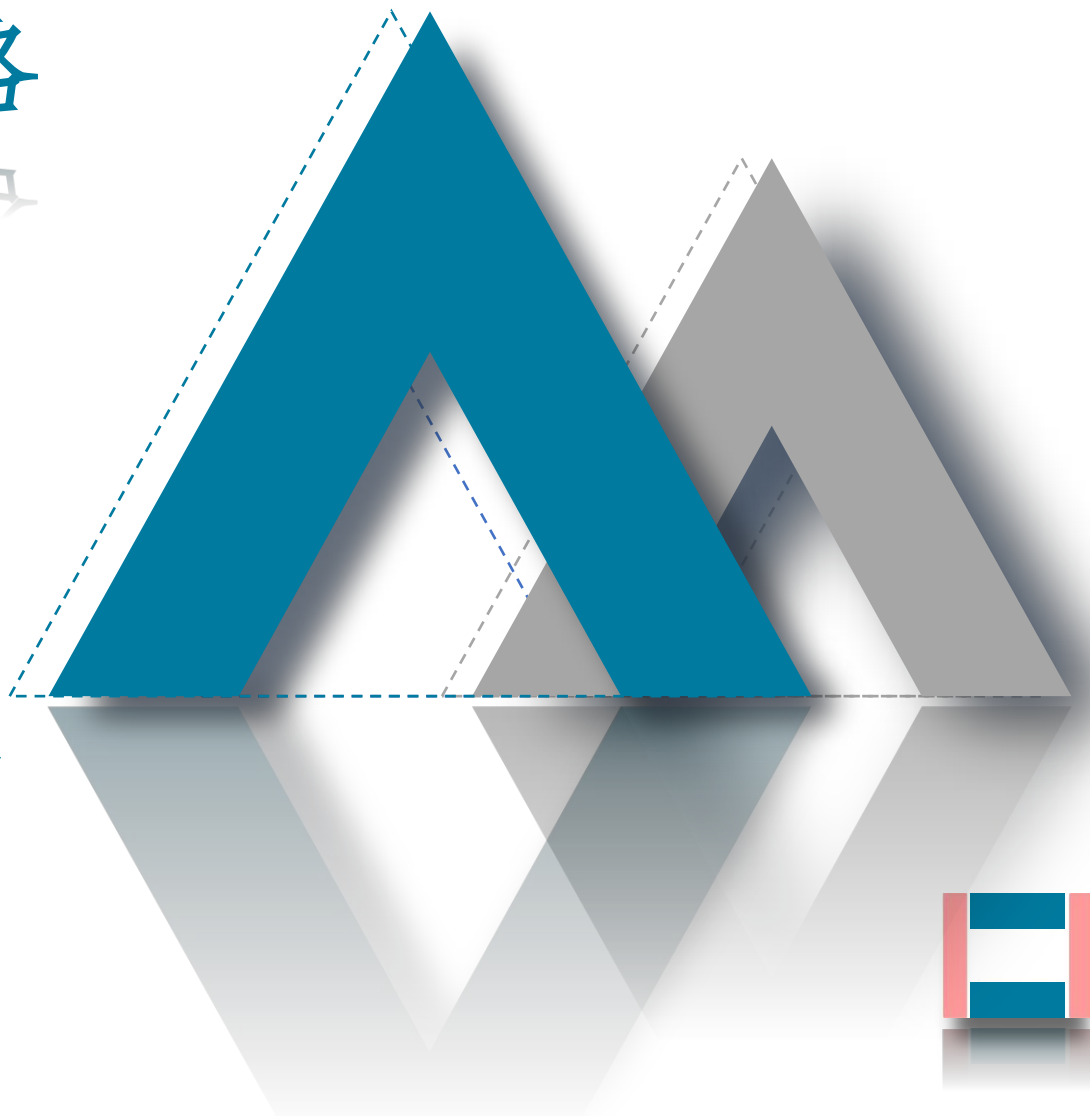
二阶动态电路

一阶动态电路

第8章

主讲人：邹建龙

时 间：2025 年 4 月 7 日



□ 引言

□ 8.1 零输入 RLC 串联二阶电路的微分方程

□ 8.2 零输入 RLC 串联二阶电路微分方程的求解

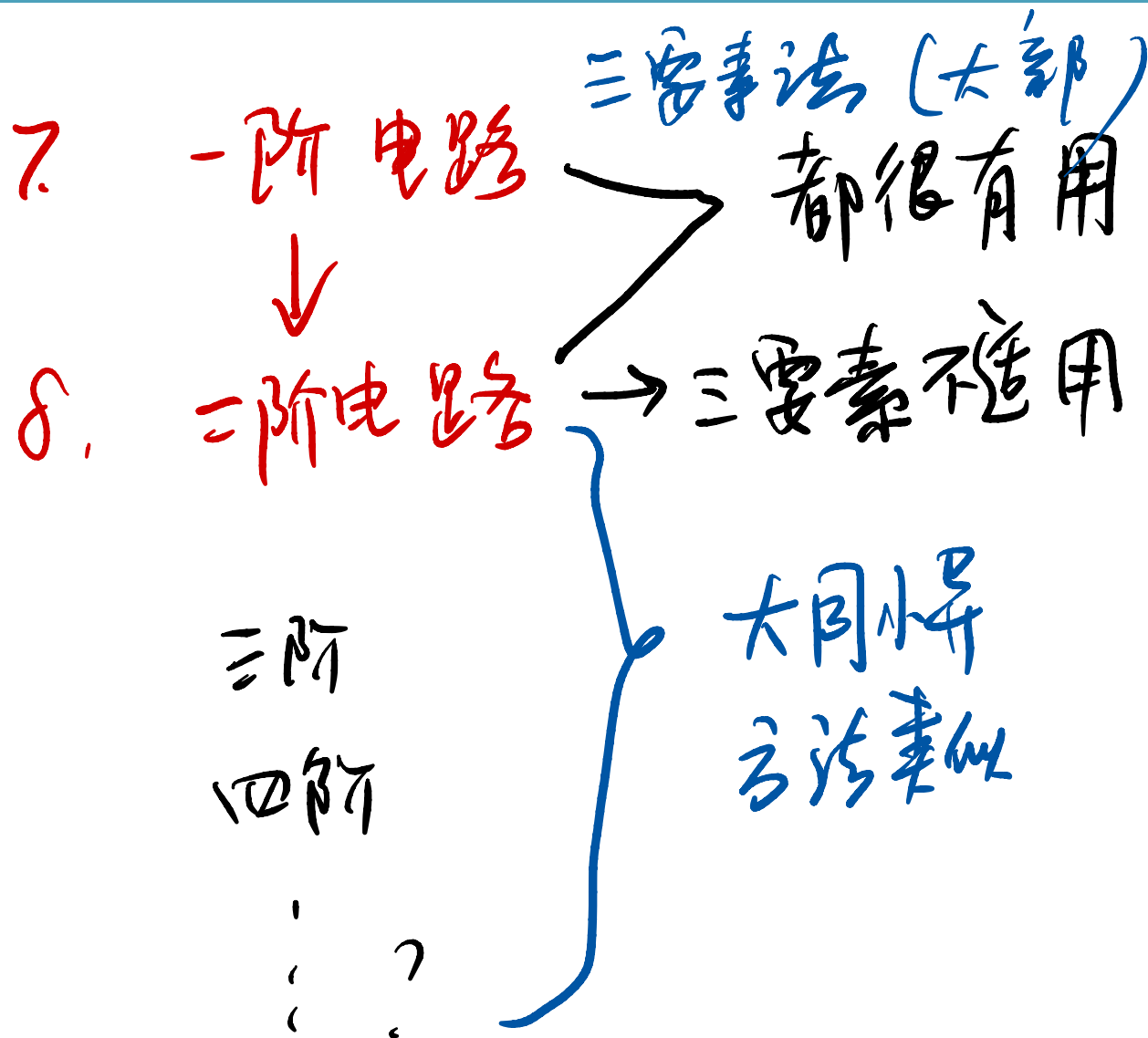
□ 8.3 零输入 RLC 串联二阶电路的工作状态

□ 8.4 其他二阶电路

□ 小结



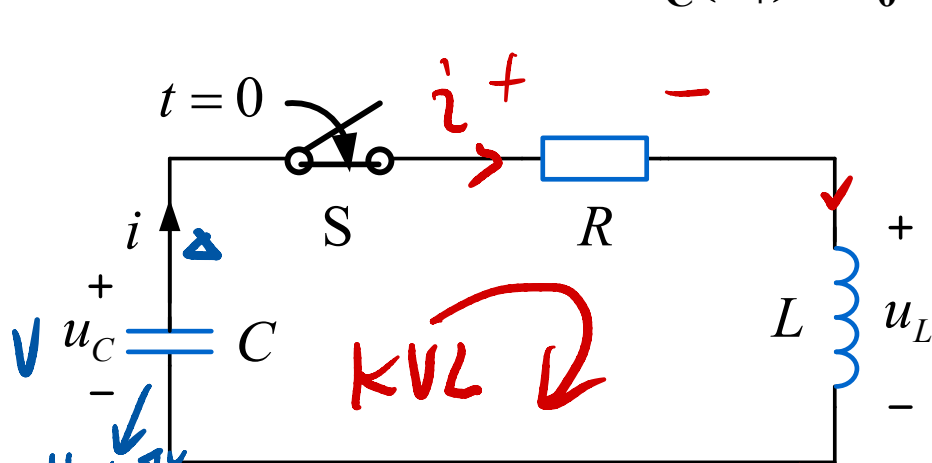
8 二阶动态电路——引言



8.1 零输入RLC串联二阶电路的微分方程

零输入RLC串联二阶电路

设电容有初始储能 $u_C(0_+) = U_0$, 电感无初始储能 $i(0_+) = 0$



$$i = -C \frac{du_C}{dt}$$

$$y = u_C$$

非关联

$$-u_C + Ri + L \frac{di}{dt} = 0 \Rightarrow LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

二阶
微分方程

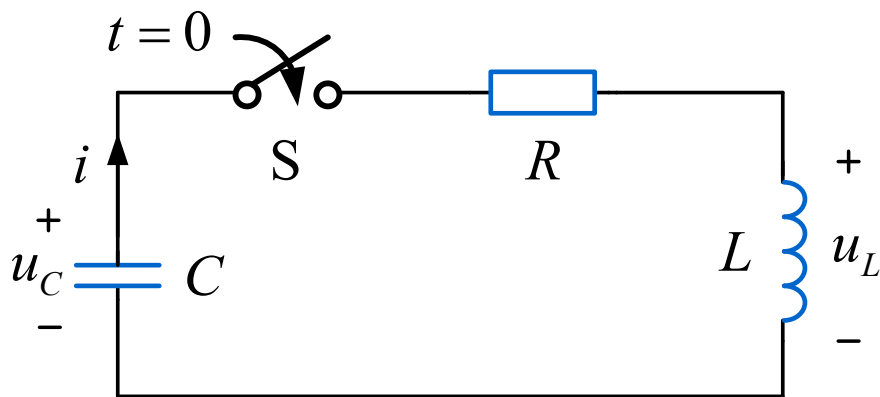
$$ay'' + by' + y = 0?$$

LC RC 1

8.1 零输入RLC串联二阶电路的微分方程

零输入RLC串联二阶电路 $u_d(0^-)=U_0$ $u_d(0^+)=u_d(0^-)=U_0$

设电容有初始储能 $u_C(0_+)=U_0$, 电感无初始储能 $i(0_+)=0$



$$-u_C + Ri + u_L = 0$$

$$i = -C \frac{du_C}{dt}$$

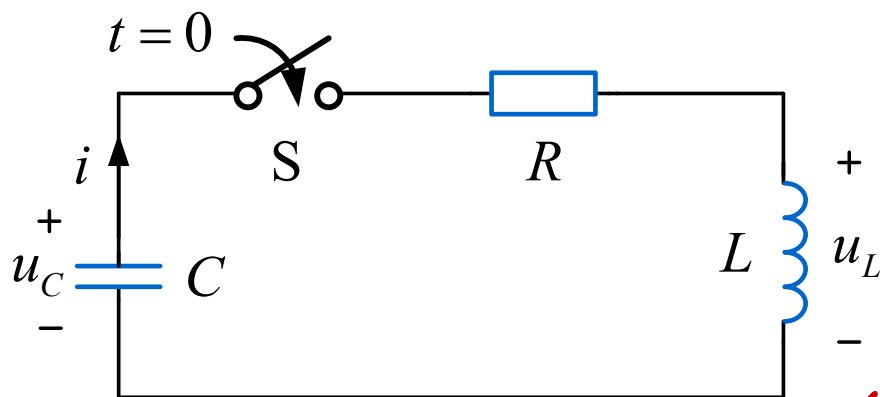
$$u_L = L \frac{di}{dt}$$

$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$LC \frac{d^2 i}{dt^2} + RC \frac{di}{dt} + i = 0$$



8.2 零输入RLC串联二阶电路微分方程的求解



$y(0)$ $y'(0)$
初始条件

$$u_C = Ae^{\lambda t}$$

$$LC \left(\frac{d^2 u_C}{dt^2} \right) + RC \left(\frac{du_C}{dt} \right) + u_C = 0$$

λ^2 λ 1

$$LC \lambda^2 + RC \lambda + 1 = 0$$

$$ax^2 + bx + c = 0$$
$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

二阶齐次微分方程的求解步骤

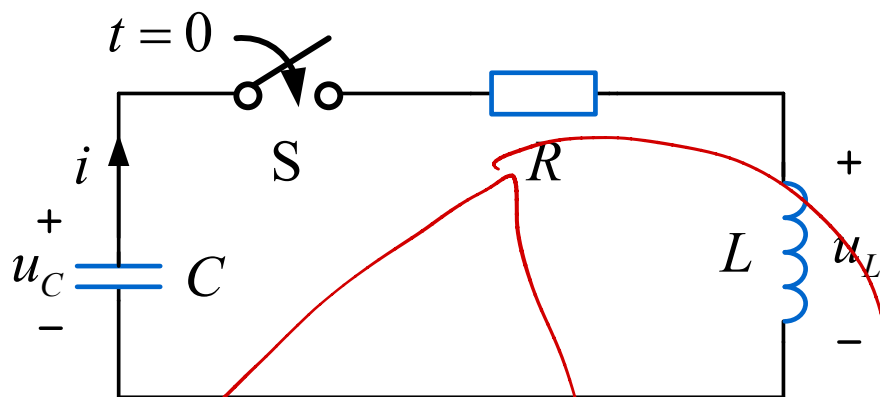
第1步：写出微分方程的**特征方程**

第2步：根据特征方程求出**特征根**

第3步：根据特征根给出**齐次通解**（含两个待定系数）

第4步：代入**初始条件**，求出两个待定系数，求解完成。

8.2 零输入RLC串联二阶电路微分方程的求解

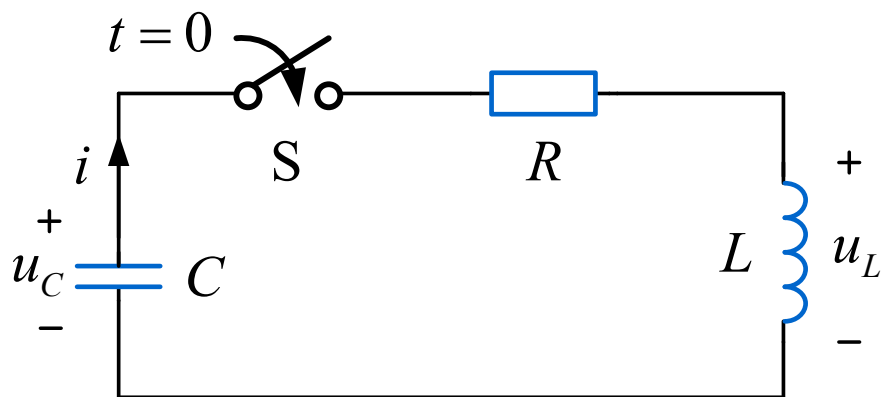


$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$LC\lambda^2 + RC\lambda + 1 = 0$$

$$\lambda = \frac{-RC \pm \sqrt{(RC)^2 - 4LC}}{2LC} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$$

8.2 零输入RLC串联二阶电路微分方程的求解



$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$LC\lambda^2 + RC\lambda + 1 = 0$$

根据特征方程根的判别式，
特征根有4种可能的情况：

(1) 两个不相等的实根

(2) 两个相等的实根

(3) 两个共轭的复根

(4) 两个共轭的纯虚根

一种

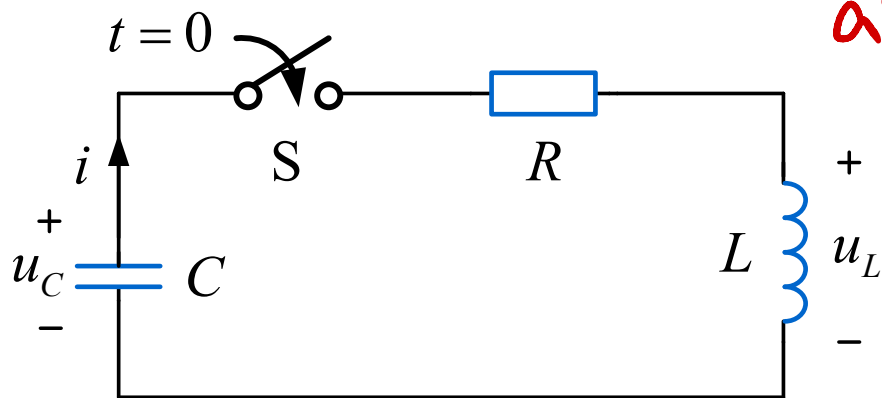
情况

(4) 是 (3) 的特例

$$\lambda = \frac{-RC \pm \sqrt{(RC)^2 - 4LC}}{2LC} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$$



8.2 零输入RLC串联二阶电路微分方程的求解



$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$LC\lambda^2 + RC\lambda + 1 = 0$$

$$ax^2 + bx + c = 0 \quad \Delta = b^2 - 4ac$$

(1) 两个不相等的实根

$$\text{需满足 } \Delta = (RC)^2 - 4LC > 0,$$

$$\text{即 } R > 2\sqrt{\frac{L}{C}}$$

根判别式

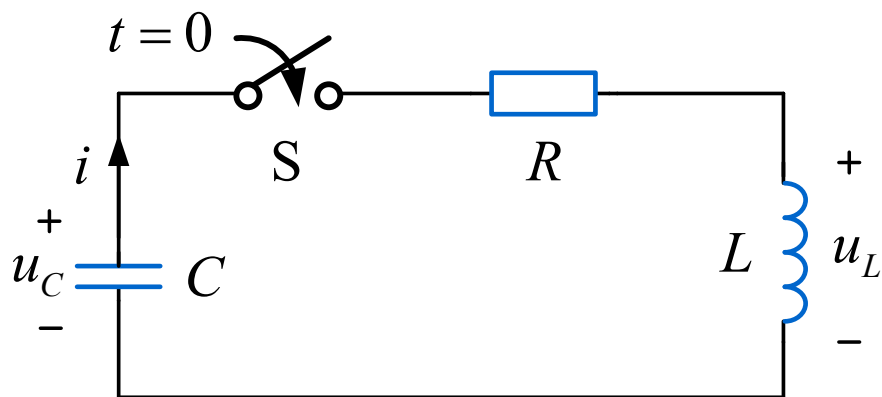
$$u_C(t) = A_1 e^{\lambda_1 t} + B_1 e^{\lambda_2 t}$$

背下来

齐次通解

$$\lambda_{1,2} = \frac{-RC \pm \sqrt{(RC)^2 - 4LC}}{2LC} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$$

8.2 零输入RLC串联二阶电路微分方程的求解



$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$LC\lambda^2 + RC\lambda + 1 = 0$$

$= 0$

$$\lambda_{1,2} = \frac{-RC \pm \sqrt{(RC)^2 - 4LC}}{2LC} = \frac{-R}{2L}$$

(2) 两个相等的实根

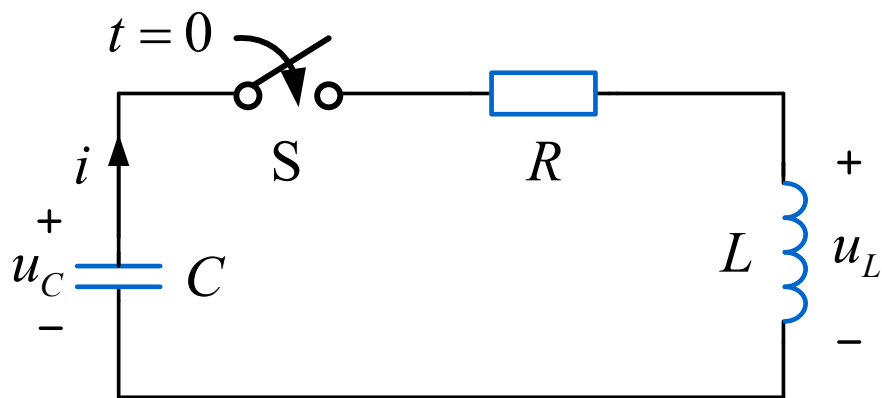
需满足 $\Delta = (RC)^2 - 4LC = 0$,

$$\text{即 } R = 2\sqrt{\frac{L}{C}}$$

$$u_C(t) = (A_2 + B_2 t) e^{\lambda t}$$

记住★

8.2 零输入RLC串联二阶电路微分方程的求解



$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$LC\lambda^2 + RC\lambda + 1 = 0$$

$$\lambda = \frac{-RC \pm \sqrt{(RC)^2 - 4LC}}{2LC} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L} \Rightarrow \alpha = \frac{R}{2L} > 0$$

$\lambda_1 = -\alpha + j\omega, \lambda_2 = -\alpha - j\omega$

$= -\alpha \pm j\omega$

$$z = a \pm ib = a \pm bi$$

(3) 两个共轭的复根

$$z = a \pm jb$$

需满足 $\Delta = (RC)^2 - 4LC < 0$,

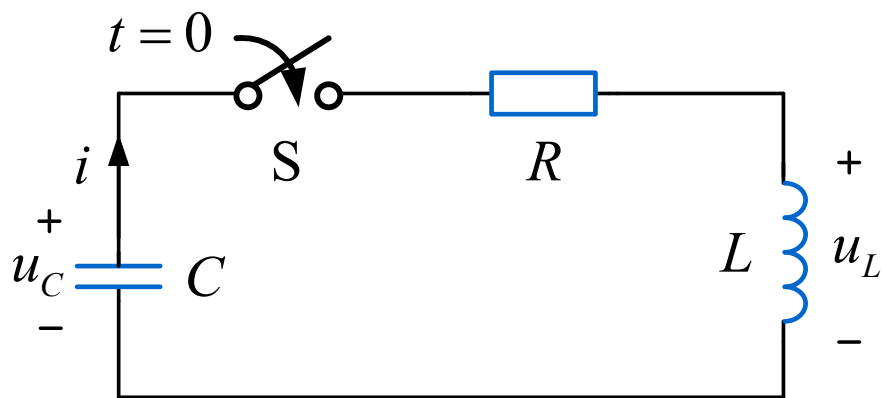
$$\text{即 } R < 2\sqrt{\frac{L}{C}}, \text{ 且 } R \neq 0$$

记住★

$$u_C(t) = e^{-\alpha t} (A_3 \cos \omega t + B_3 \sin \omega t)$$



8.2 零输入RLC串联二阶电路微分方程的求解



(4) 两个共轭的纯虚根

需满足 $R=0$

$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$LC\lambda^2 + RC\lambda + 1 = 0$$

$$\lambda = \frac{-RC \pm \sqrt{(RC)^2 - 4LC}}{2LC} = \frac{\pm \sqrt{-\frac{4L}{C}}}{2L}$$

$$\lambda_1 = +j\omega, \quad \lambda_2 = -j\omega$$

$$u_C(t) = A_4 \cos \omega t + B_4 \sin \omega t$$

$e^{-\alpha t}$
 $\alpha = 0$

记住

$\Rightarrow \alpha = 0$

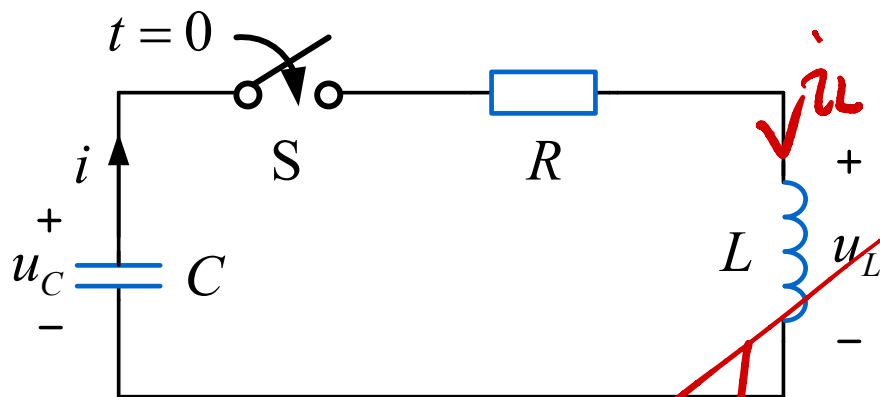
8.2 零输入RLC串联二阶电路微分方程的求解

齐次二阶微分方程解的类型总结

判断条件	特征根	解的表达式
过 $R > 2\sqrt{\frac{L}{C}}$ $\Delta > 0$	$\lambda_{1,2} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$ 不相等实根	$u_C(t) = A_1 e^{\lambda_1 t} + B_1 e^{\lambda_2 t}$
临 $R = 2\sqrt{\frac{L}{C}}$ $\Delta = 0$	$\lambda = \lambda_{1,2} = \frac{-R}{2L}$ 相等实根	$u_C(t) = (A_2 + B_2 t) e^{\lambda t}$
欠 $0 < R < 2\sqrt{\frac{L}{C}}$ $\Delta < 0$	$\lambda_{1,2} = \frac{-R \pm j\sqrt{\frac{4L}{C} - R^2}}{2L}$ $= -\alpha \pm j\omega$ 共轭虚根	$u_C(t) = e^{-\alpha t} (A_3 \cos \omega t + B_3 \sin \omega t)$
$R = 0$	$\lambda_{1,2} = \frac{\pm j\sqrt{\frac{4L}{C}}}{2L} = \pm j\sqrt{\frac{1}{LC}} = \pm j\omega$	$u_C(t) = A_4 \cos \omega t + B_4 \sin \omega t$

8.2 零输入RLC串联二阶电路微分方程的求解

第4步：最后一个求解步骤，代入初始条件，确定待定系数
(以两个不相等实根为例)



$$u_C(t) = \begin{cases} Ae^{\lambda_1 t} + Be^{\lambda_2 t} \\ (A+Bt)e^{\lambda t} \\ e^{-\alpha t}(A\cos\omega t + B\sin\omega t) \end{cases}$$

A, B 待定系数

$$u_C(0_-) = U_0$$

$$u_C(0_+) = u_C(0_-) = U_0$$

$$u_C'(0_+) = \left. \frac{du_C}{dt} \right|_{t=0_+} ?$$

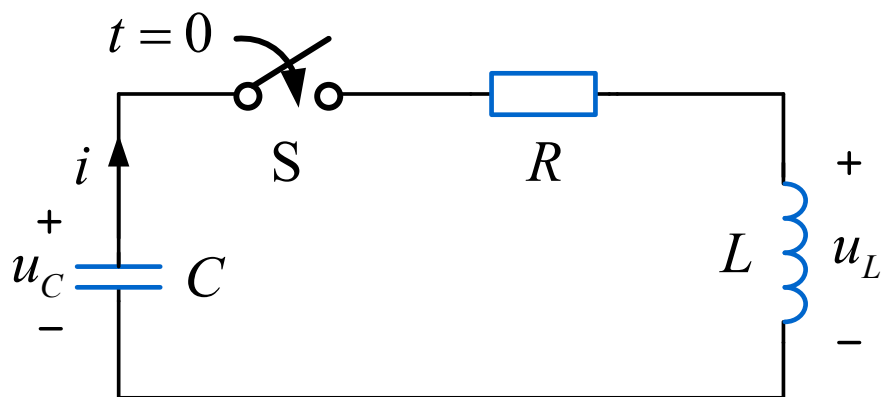
$$i_L(0_+) = i(0_+) = i(0_-) = 0 \text{ A}$$

$$i_C = -C \frac{du_C}{dt}$$

$$\frac{du_C}{dt} = -\frac{1}{C} i_C \Rightarrow \left. \frac{du_C}{dt} \right|_{t=0_+} = 0$$

8.2 零输入RLC串联二阶电路微分方程的求解

第4步：最后一个求解步骤，代入初始条件，确定待定系数
(以两个不相等实根为例)



$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$\lambda_{1,2} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$$

(1) 两个不相等的实根

$$u_C(t) = A_1 e^{\lambda_1 t} + B_1 e^{\lambda_2 t}$$

$$u_C(0_+) = u_C(0_-) = U_0$$

$$U_0 = A_1 + B_1$$

$$u_C(0_+) = A_1 e^0 + B_1 e^0$$

$$U_0 = A + B$$

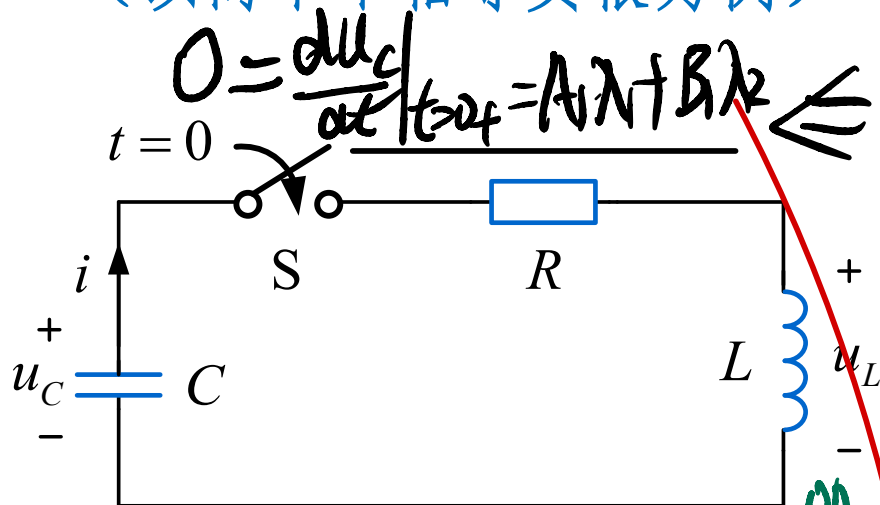
$$e^0 = 1$$



8.2 零输入RLC串联二阶电路微分方程的求解

第4步：最后一个求解步骤，代入初始条件，确定待定系数

(以两个不相等实根为例)



$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$\lambda_{1,2} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$$

(1) 两个不相等的实根 λ_1, λ_2

$$\frac{du_C}{dt} = A_1 \lambda_1 e^{\lambda_1 t} + B_1 \lambda_2 e^{\lambda_2 t}$$
$$u_C(t) = A_1 e^{\lambda_1 t} + B_1 e^{\lambda_2 t}$$
$$\frac{du_C(t)}{dt} = A_1 \lambda_1 e^{\lambda_1 t} + B_1 \lambda_2 e^{\lambda_2 t}$$

② $i(0_+) = 0 = -C \left[\frac{du_C}{dt} \right]_{t=0_+}$

$$0 = A_1 \lambda_1 + B_1 \lambda_2$$
$$U_0 = A_1 + B_1$$

①

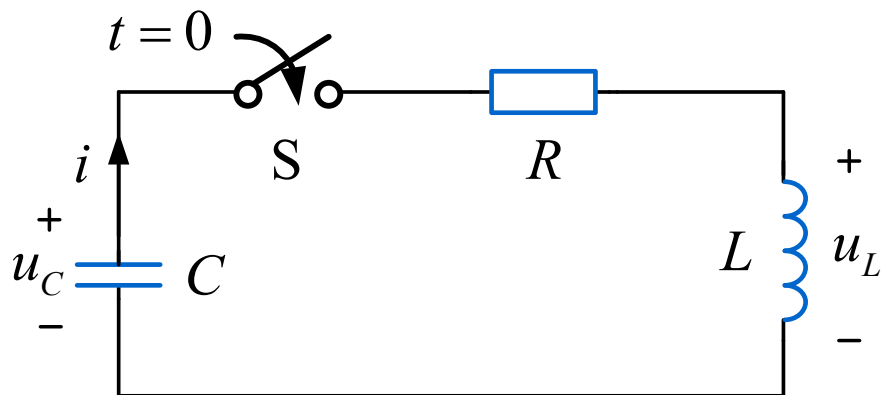
$$A_1 = \frac{-\lambda_2 U_0}{\lambda_1 - \lambda_2}, \quad B_1 = \frac{\lambda_1 U_0}{\lambda_1 - \lambda_2}$$

② $\frac{du_C}{dt} \big|_{t=0_+} = 0$

8.2 零输入RLC串联二阶电路微分方程的求解

第4步：最后一个求解步骤，**代入初始条件，确定待定系数**
(以两个不相等实根为例)

(1) 两个**不相等的实根**



$$\lambda_{1,2} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$$

< R

↓

$$\lambda_1 < 0$$
$$\lambda_2 < 0$$

R > 0

$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$u_C(t) = A_1 e^{\lambda_1 t} + B_1 e^{\lambda_2 t}$$

$$A_1 = \frac{-\lambda_2 U_0}{\lambda_1 - \lambda_2}, \quad B_1 = \frac{\lambda_1 U_0}{\lambda_1 - \lambda_2}$$

$$u_C(t) = \frac{-\lambda_2 U_0}{\lambda_1 - \lambda_2} e^{\lambda_1 t} + \frac{\lambda_1 U_0}{\lambda_1 - \lambda_2} e^{\lambda_2 t}$$

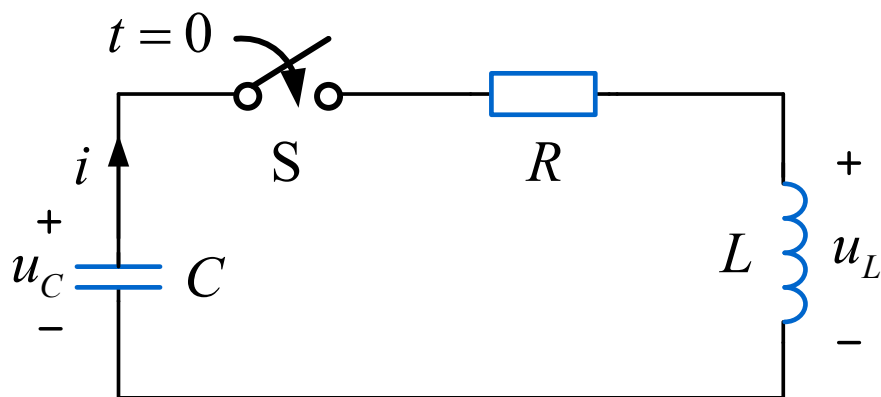
不用背



8.3 零输入RLC串联二阶电路的工作状态

零输入RLC串联二阶电路的4种工作状态

二阶微分方程的特征根的4种情况对应4种解的形式，
每种解的形式称为二阶电路的一种工作状态



$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = 0$$

$$\lambda_{1,2} = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$$

特征根有4种可能的情况：

(1) 两个不相等的实根 (过阻尼状态)

$$Ae^{\lambda_1 t} + Be^{\lambda_2 t}$$

R 大

(2) 两个相等的实根 (临界阻尼状态)

$$R > 2\sqrt{L/C}$$

$$R = 2\sqrt{L/C}$$

$$(A+Bt)e^{\lambda t}$$

(3) 两个共轭的复根 (欠阻尼状态)

$$e^{-\alpha t} (A \cos \omega t + B \sin \omega t)$$

$$0 < R < 2\sqrt{L/C}$$

(4) 两个共轭的纯虚根 (零阻尼状态)

$$A \cos \omega t + B \sin \omega t$$

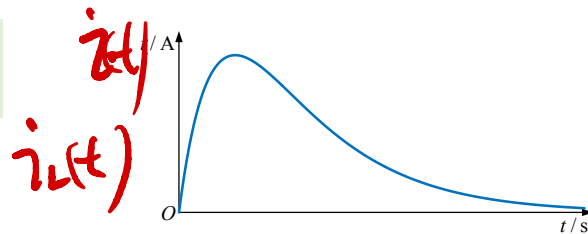
$$R = 0$$

8.3 零输入RLC串联二阶电路的工作状态

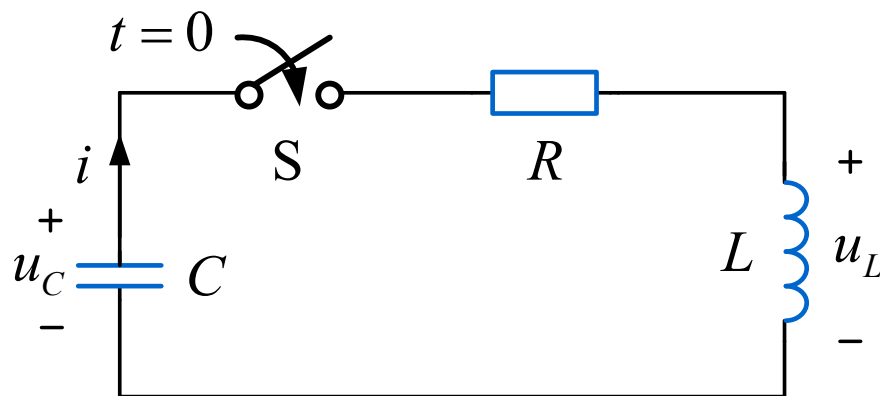
工作状态1: 过阻尼

$$R > 2\sqrt{L/C}$$

典型特征: 电容电压单调衰减。

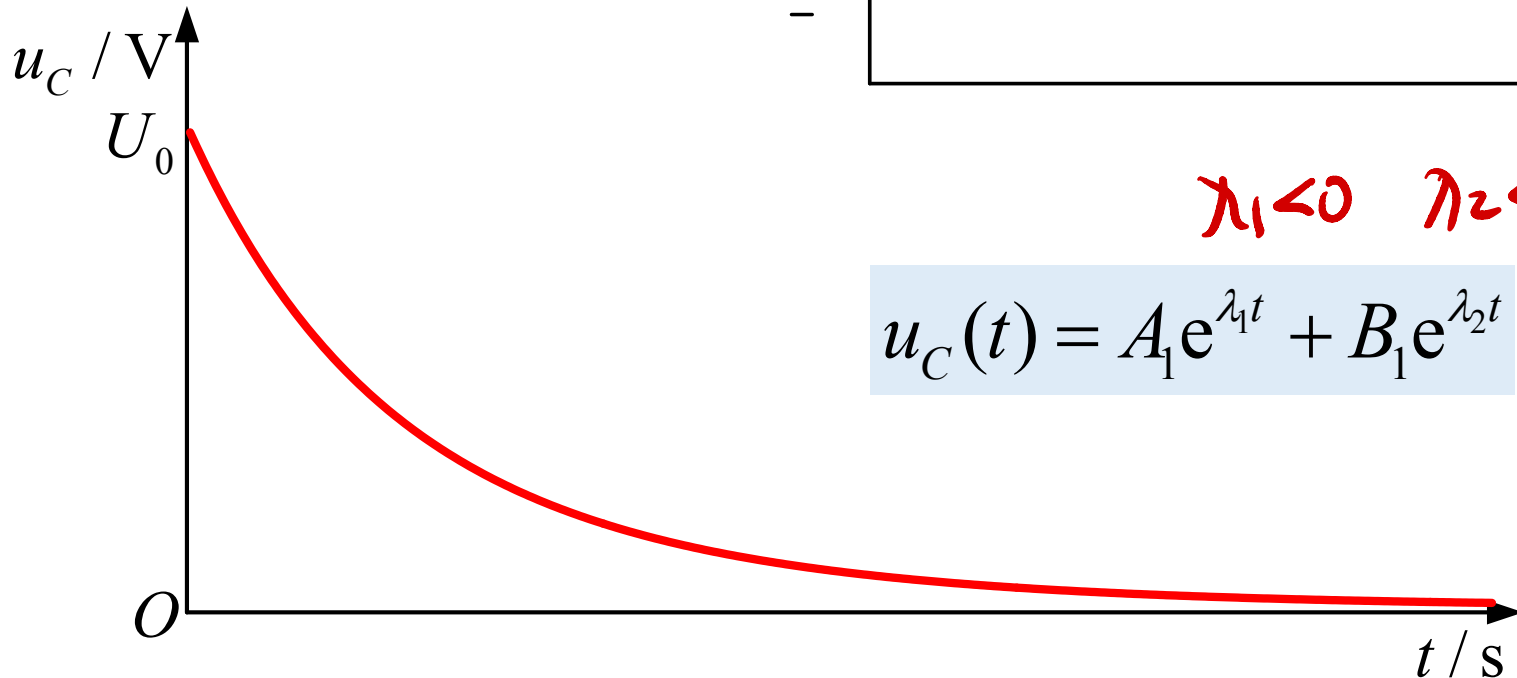


$$\Delta > 0$$



$$\lambda_1 < 0 \quad \lambda_2 < 0$$

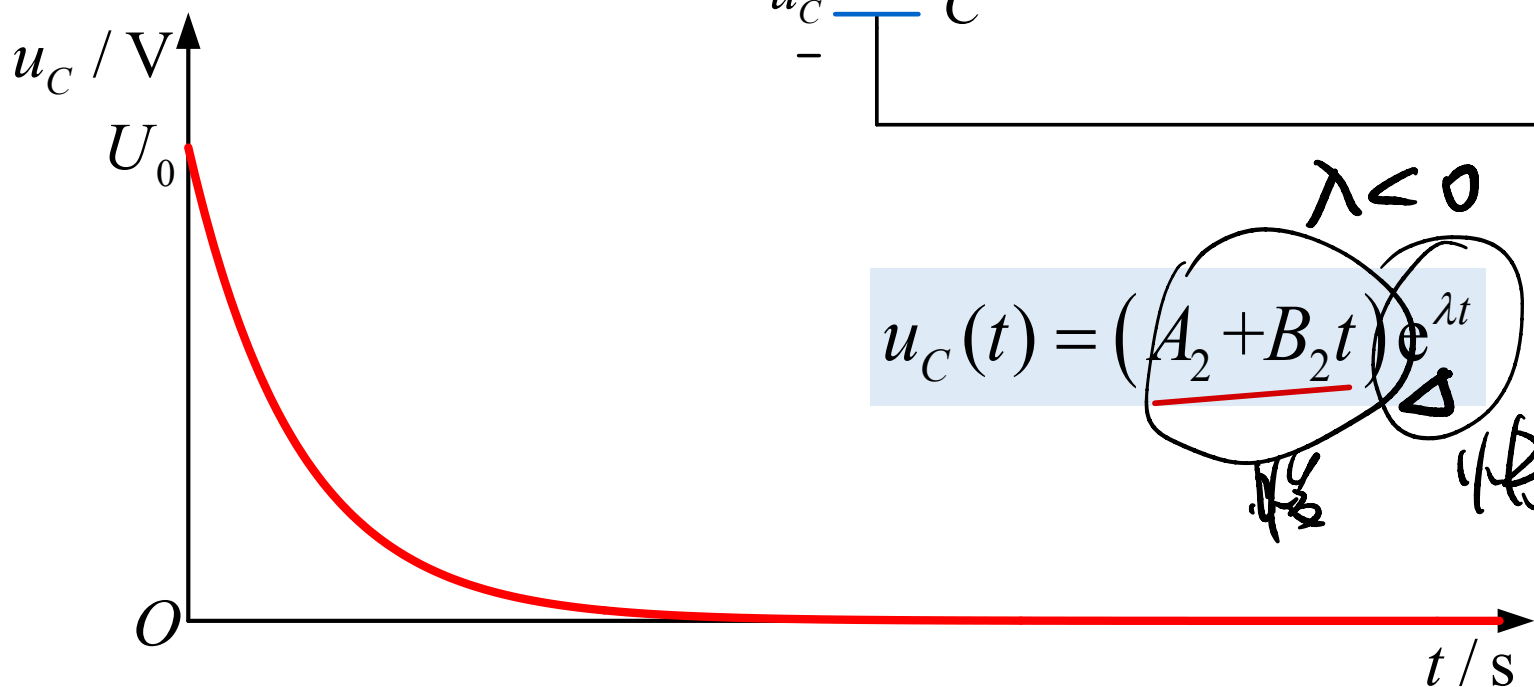
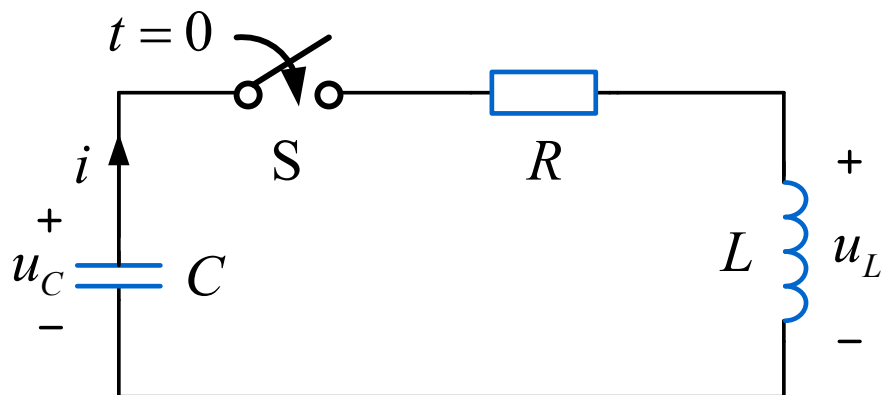
$$u_C(t) = A_1 e^{\lambda_1 t} + B_1 e^{\lambda_2 t}$$



8.3 零输入 RLC 串联二阶电路的工作状态

工作状态2：临界阻尼 $R = 2\sqrt{L/C}$

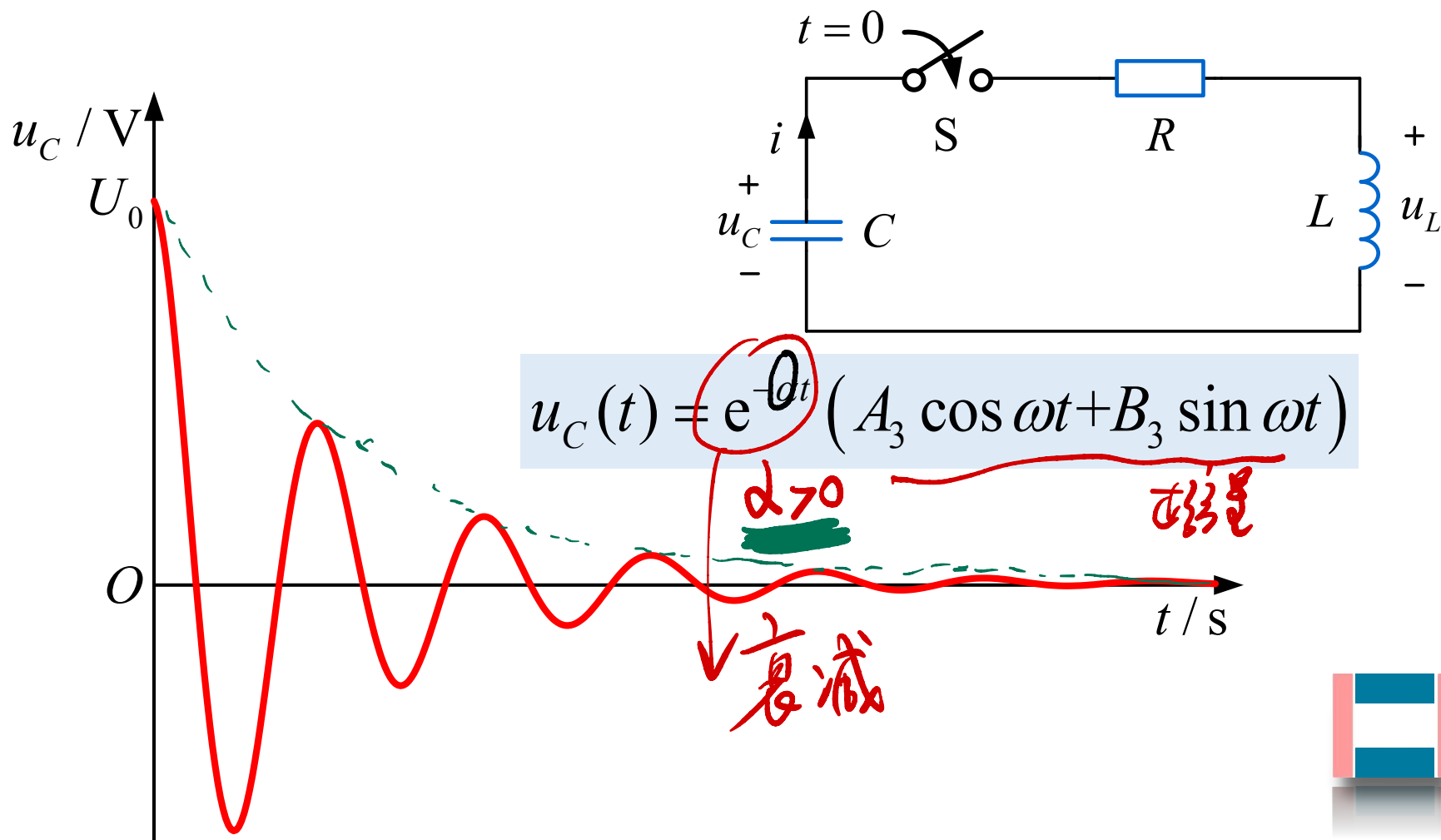
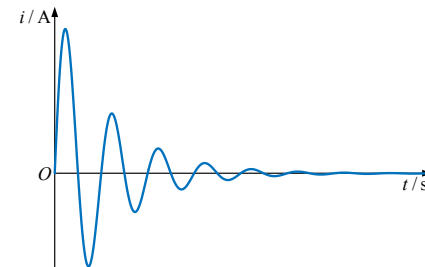
典型特征：电容电压单调衰减，比过阻尼衰减更快。



8.3 零输入RLC串联二阶电路的工作状态

工作状态3：欠阻尼 $0 < R < 2\sqrt{L/C}$

典型特征：电容电压振荡衰减。



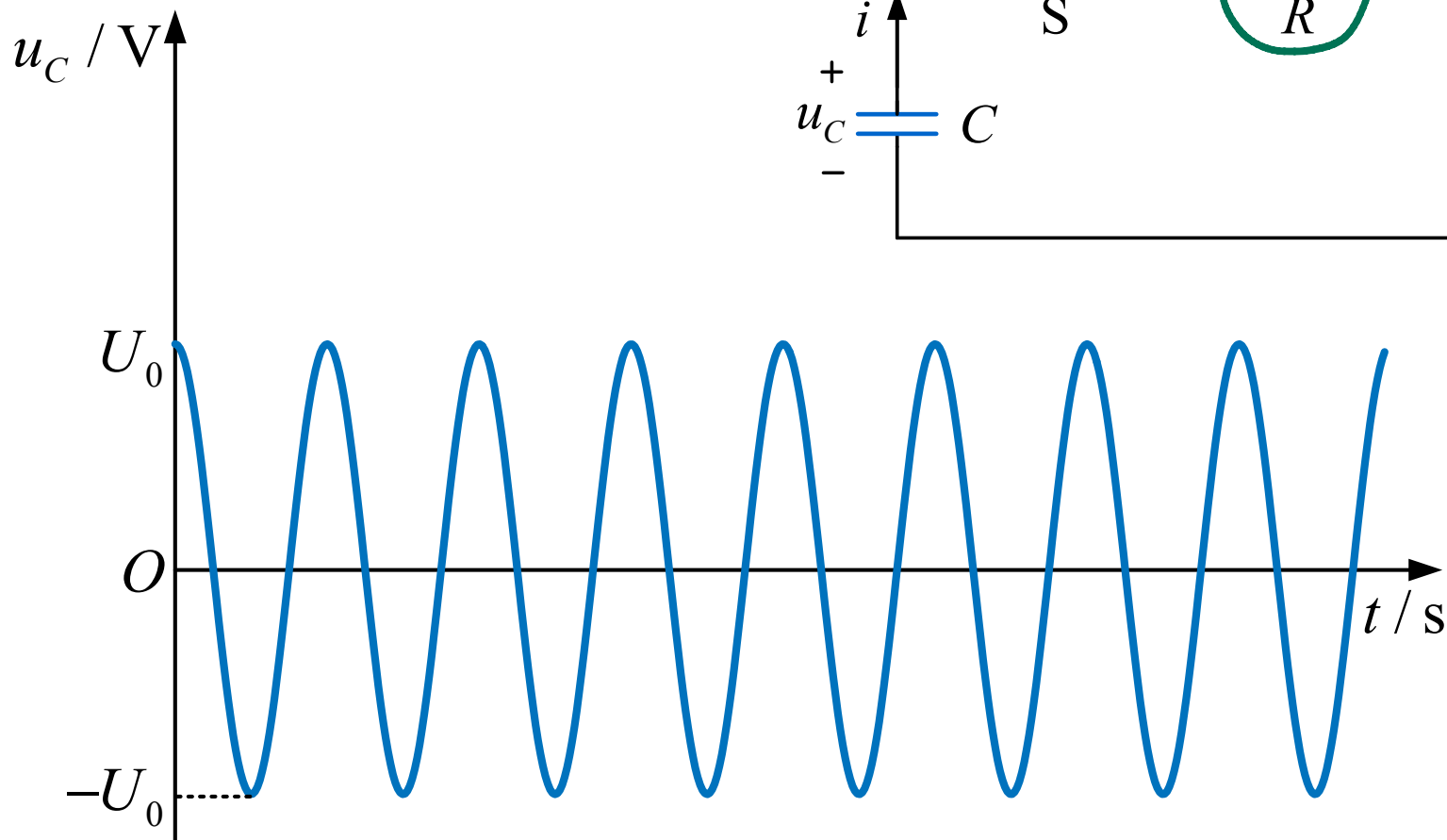
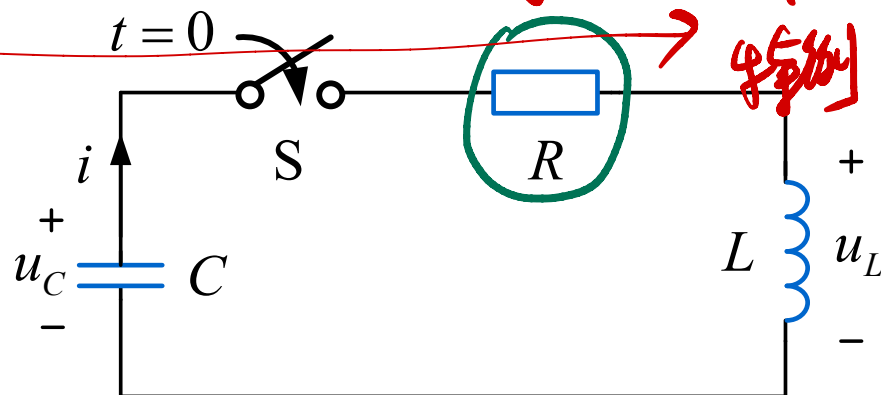
8.3 零输入RLC串联二阶电路的工作状态

工作状态4: 零阻尼 $R = 0$

典型特征: 电容电压为不衰减的正弦量。

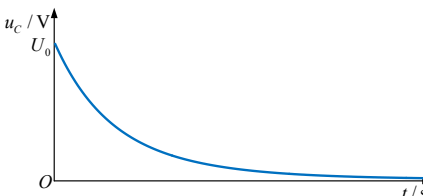
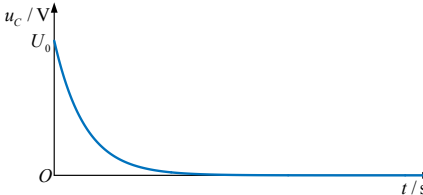
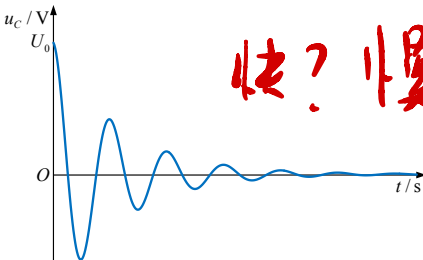
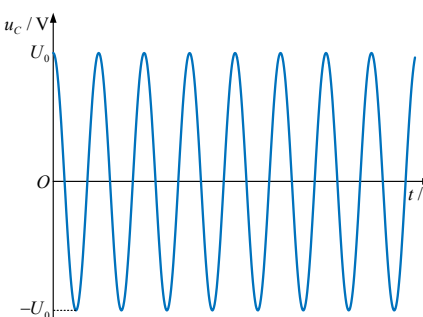
$$u_C(t) = A_4 \cos \omega t + B_4 \sin \omega t$$

① 过阻尼
② 临界阻尼
③ 欠阻尼
特例



8.3 零输入RLC串联二阶电路的工作状态

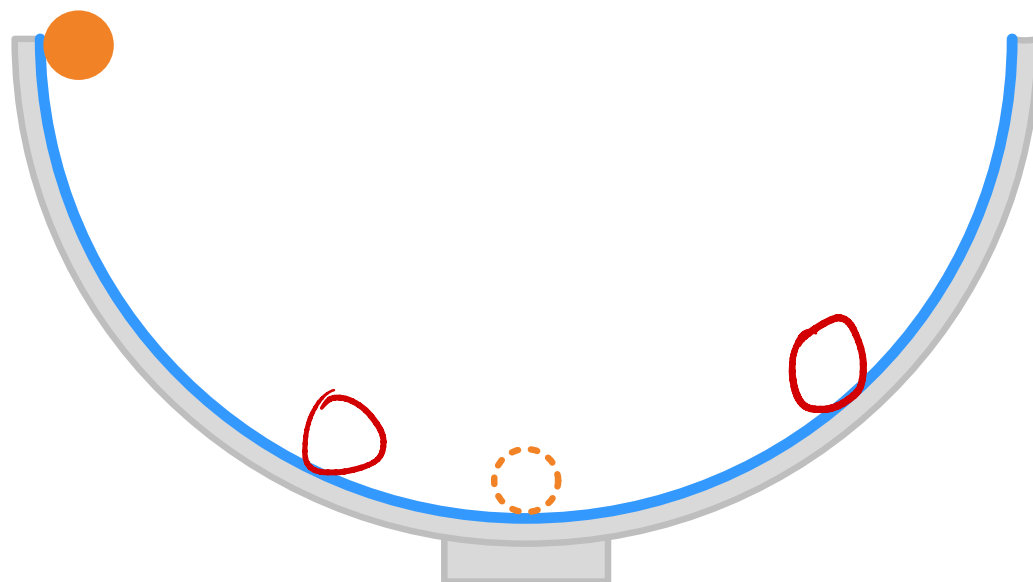
零输入RLC串联二阶电路4种工作状态总结

判断条件	工作状态	电容电压波形	动态响应特点
$R > 2\sqrt{\frac{L}{C}}$	过阻尼		单调衰减
$R = 2\sqrt{\frac{L}{C}}$	临界阻尼		单调衰减， 但衰减较之过阻尼更快
$0 < R < 2\sqrt{\frac{L}{C}}$	欠阻尼		振荡衰减
$R = 0$	零阻尼		等幅振荡，不衰减



8.3 零输入 RLC 串联二阶电路的工作状态

零输入 RLC 串联二阶电路4种工作状态的日常生活类比

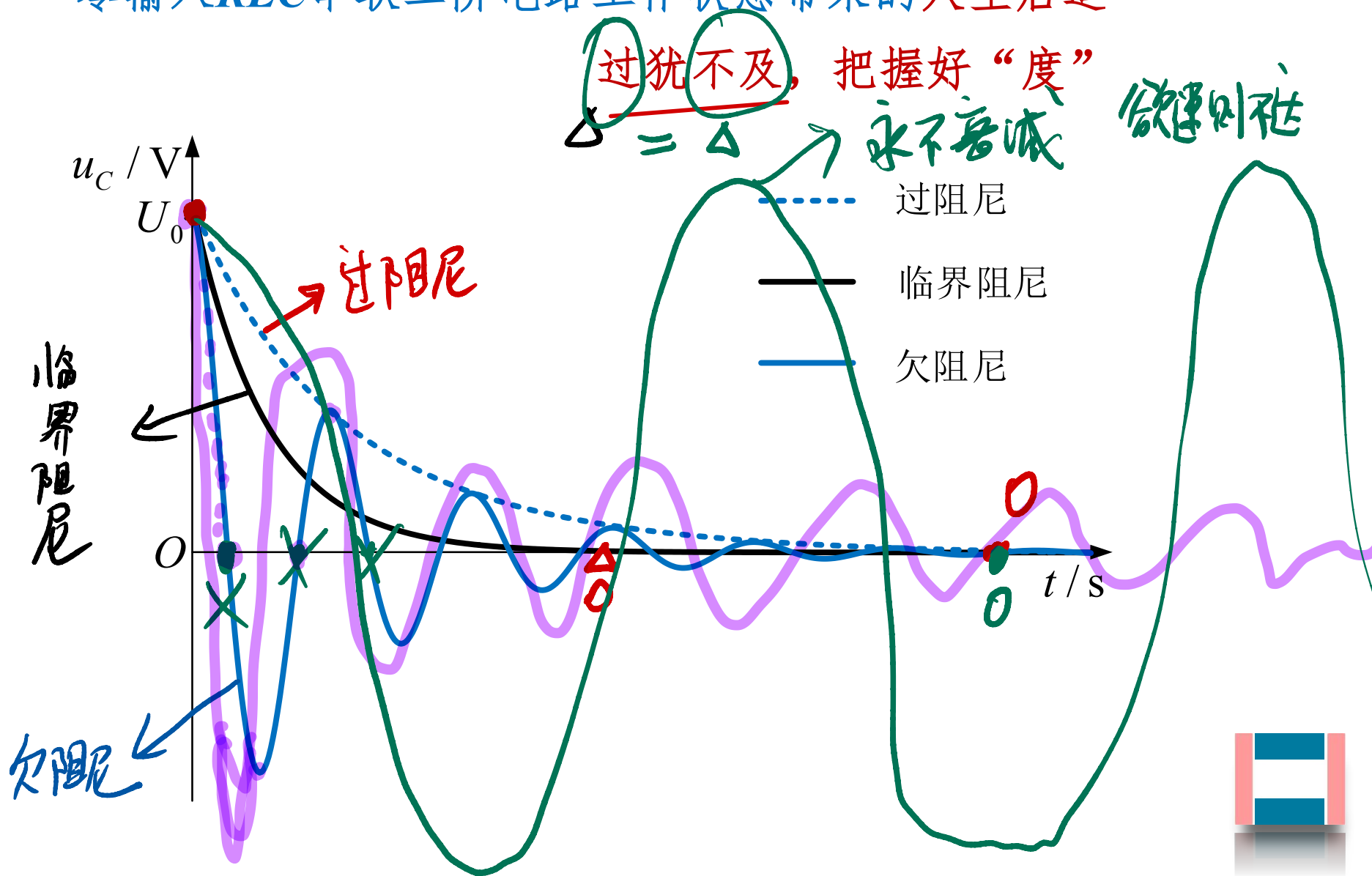


欠阻尼



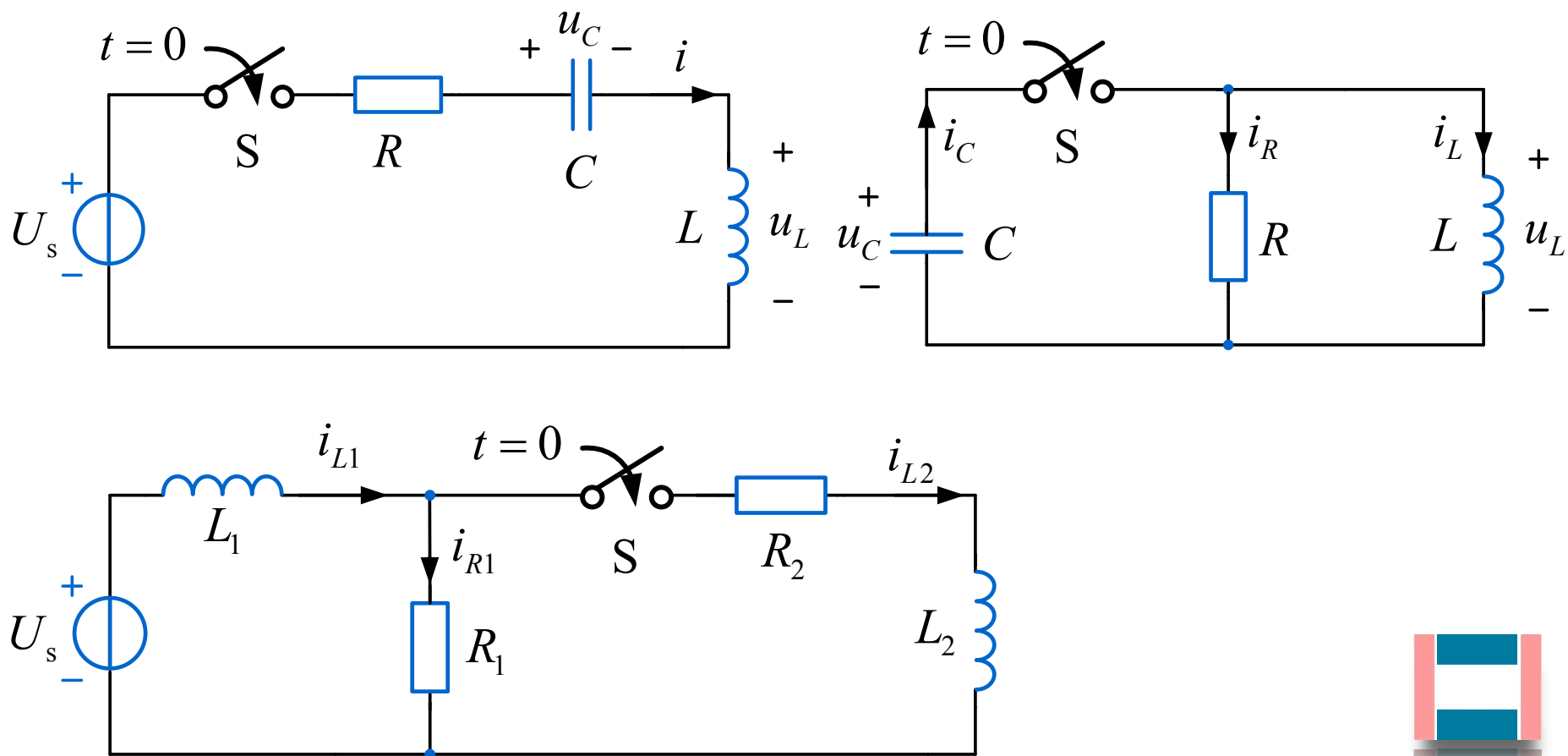
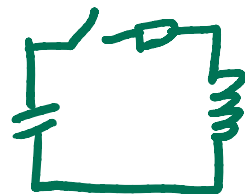
8.3 零输入 RLC 串联二阶电路的工作状态

零输入 RLC 串联二阶电路工作状态带来的人生启迪



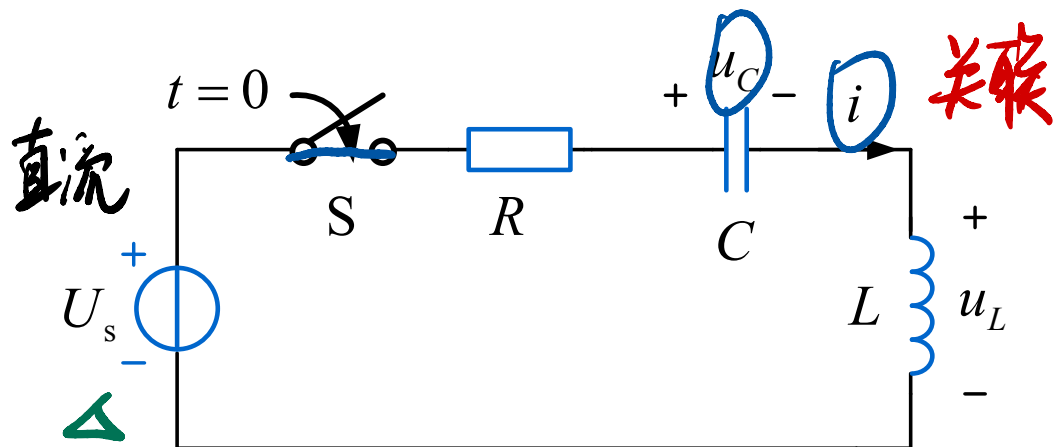
8.4 其他二阶电路

- 不是零输入 RLC 串联二阶电路的二阶电路称为其他二阶电路
- 其他二阶电路的求解过程与 RLC 串联二阶电路类似
- 其他二阶电路的工作状态与 RLC 串联二阶电路类似



8.4 其他二阶电路

(1) 含源 RLC 串联二阶电路 (以特征根为不等实根为例)



$$-U_s + Ri + U_C + L \frac{di}{dt} = 0$$

$$i = C \frac{dU_C}{dt}$$

稳态时 U_C 不变

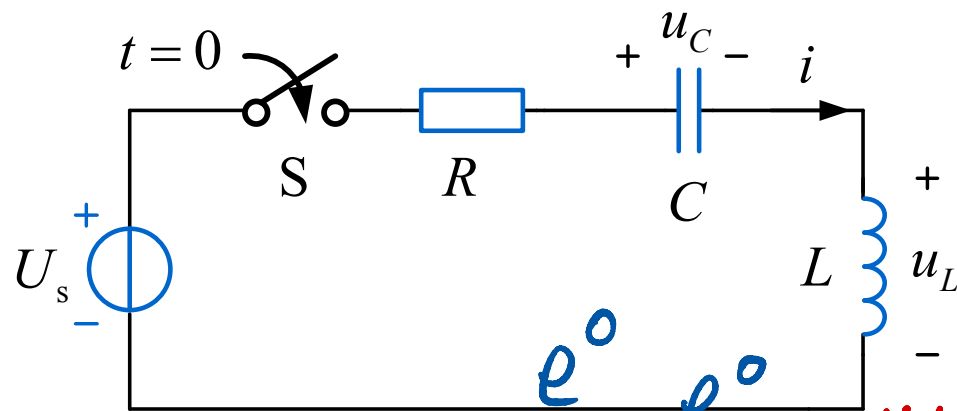
$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = U_s$$

△

非齐次

8.4 其他二阶电路

(1) 含源RLC串联二阶电路 (以特征根为不等实根为例)



$$LC \frac{d^2 u_C}{dt^2} + RC \frac{du_C}{dt} + u_C = U_s$$

$$LC\lambda^2 + RC\lambda + 1 = 0 \quad \text{齐通}$$

$$u_C(t) = u_C^{(1)}(t) + u_C^{(2)}(t)$$

$$u_C(t) = U_s + A e^{\lambda_1 t} + B e^{\lambda_2 t} \quad (3) \quad \text{非齐次特解}$$

$$0 = A\lambda_1 e^0 + B\lambda_2 e^0 \quad (4)$$

$$A = \frac{-\lambda_2}{\lambda_1 - \lambda_2} (U_0 - U_s), \quad B = \frac{\lambda_1}{\lambda_1 - \lambda_2} (U_0 - U_s)$$

A, B

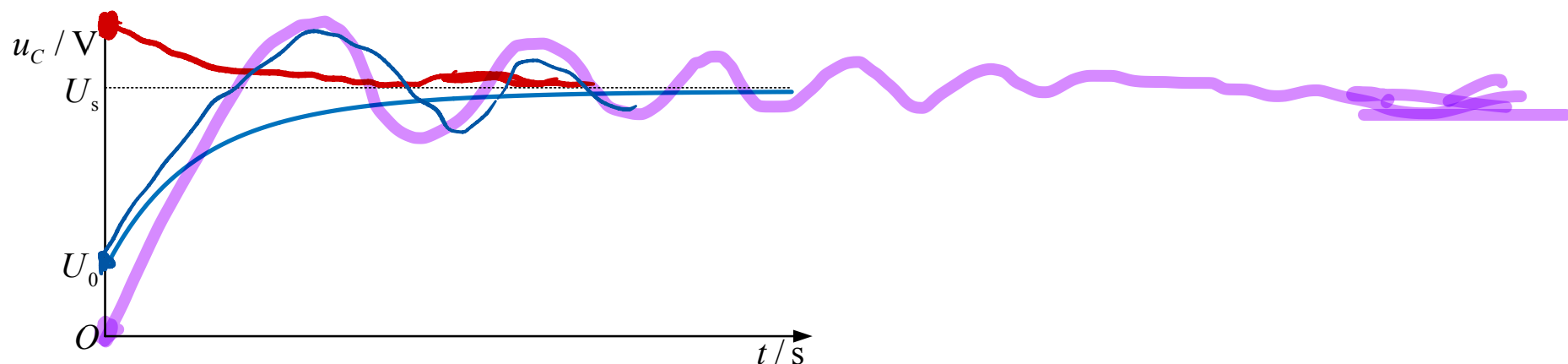
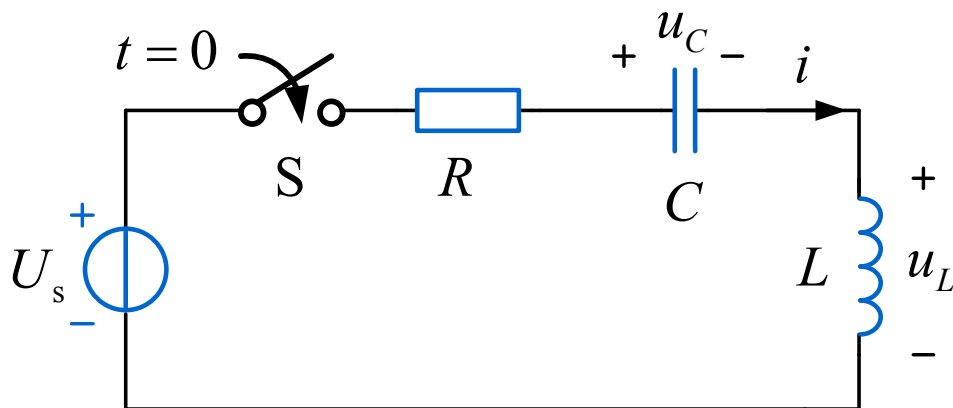
$$u_C(0_+) = U_0 \quad ? \quad ?$$

$$i(0_+) = 0 = C \left(\frac{du_C}{dt} \Big|_{t=0_+} \right)$$

$$u_C(t) = U_s + \frac{-\lambda_2}{\lambda_1 - \lambda_2} (U_0 - U_s) e^{\lambda_1 t} + \frac{\lambda_1}{\lambda_1 - \lambda_2} (U_0 - U_s) e^{\lambda_2 t}$$

8.4 其他二阶电路

(1) 含源 RLC 串联二阶电路（以特征根为不等实根为例）

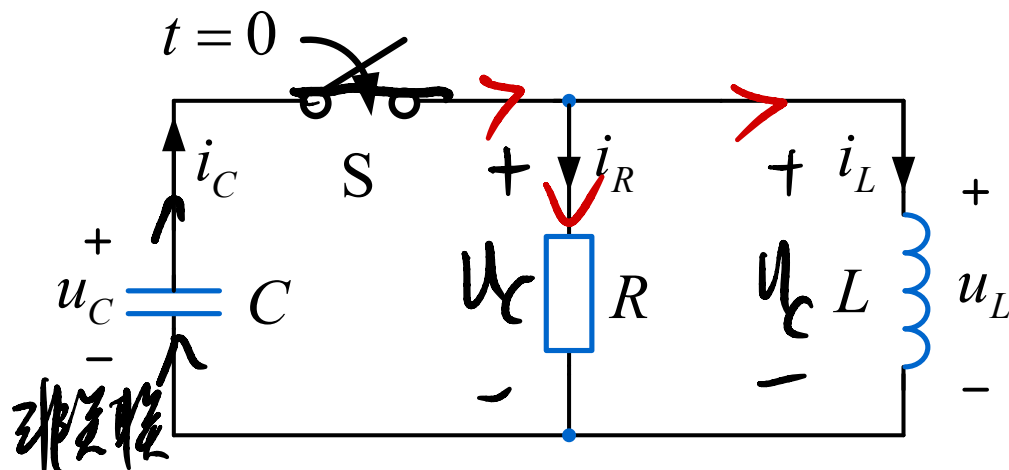


$$u_C(t) = U_s + \frac{-\lambda_2}{\lambda_1 - \lambda_2} (U_0 - U_s) e^{\lambda_1 t} + \frac{\lambda_1}{\lambda_1 - \lambda_2} (U_0 - U_s) e^{\lambda_2 t}$$



8.4 其他二阶电路

(2) 零输入RLC并联二阶电路



$$u_L = u_C = L \frac{di_L}{dt}$$
$$\frac{di_L}{dt} = \frac{1}{L} u_C$$

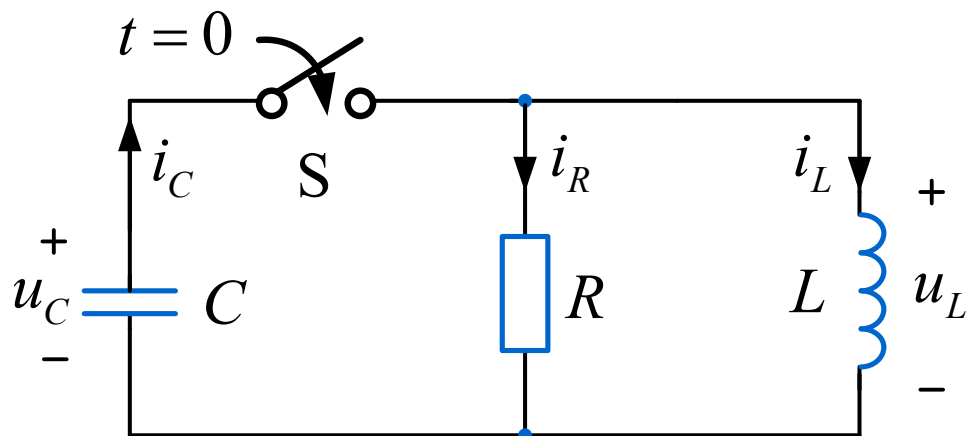
$$i_C = i_R + i_L \Rightarrow -C \frac{du_C}{dt} = \frac{u_C}{R} + i_L$$

$$-C \frac{d^2 u_C}{dt^2} = \frac{1}{R} \frac{du_C}{dt} + \frac{di_L}{dt}$$

$$\Rightarrow LC \frac{d^2 u_C}{dt^2} + \frac{L}{R} \frac{du_C}{dt} + u_C = 0$$
$$= \frac{1}{R} \frac{du_C}{dt} + \frac{1}{L} u_C$$

8.4 其他二阶电路

(2) 零输入 RLC 并联二阶电路



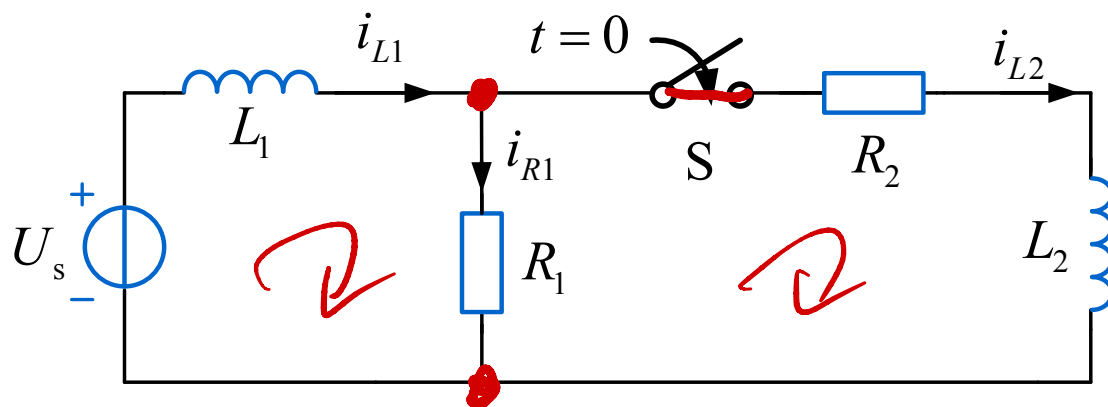
$$LC \frac{d^2 u_C}{dt^2} + \frac{L}{R} \frac{du_C}{dt} + u_C = 0$$

求解过程与 RLC 串联二阶电路极为类似，故省略



8.4 其他二阶电路

(3) 既含源，又非 RLC 串并联的二阶电路



$KCL + KVL$
(1个) (2个)

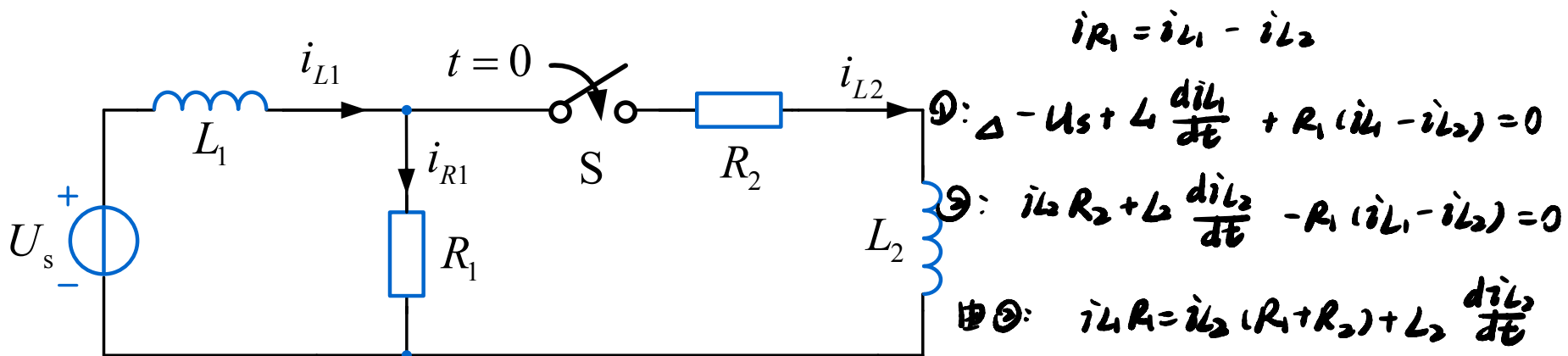
步骤:

- 列写二阶微分方程
- 写出特征方程，并求出特征根
- 根据特征根得到方程解的形式
- 代入初始条件（初值和一阶导数初值），求出待定系数



8.4 其他二阶电路

(3) 既含源，又非 RLC 串并联的二阶电路



步骤:

□ 列写二阶微分方程

□ 写出特征方程，并求出特征根

□ 根据特征根得到方程解的形式

□ 代入初始条件（初值和一阶导数初值），求出待定系数

代入①:

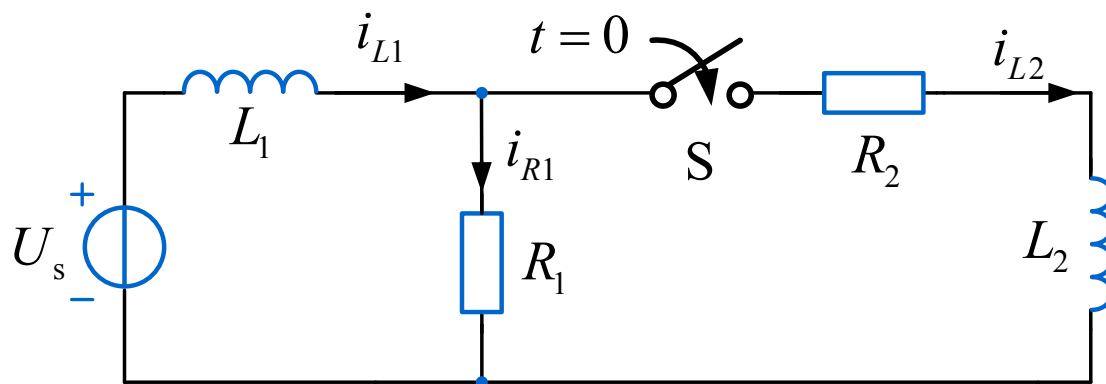
$$\frac{L_1 L_2}{R_1} \frac{d^2 i_{L2}}{dt^2} + \left[\frac{R_1 + R_2}{R_1} L_1 + L_2 \right] \frac{di_{L2}}{dt} + R_2 i_{L2} = U_s$$



8 二阶动态电路

例题1 (提高)

$$U_s = 6\text{V}, R_1 = 2\Omega, R_2 = 6\Omega, L_1 = 1\text{H}, L_2 = 4\text{H}$$



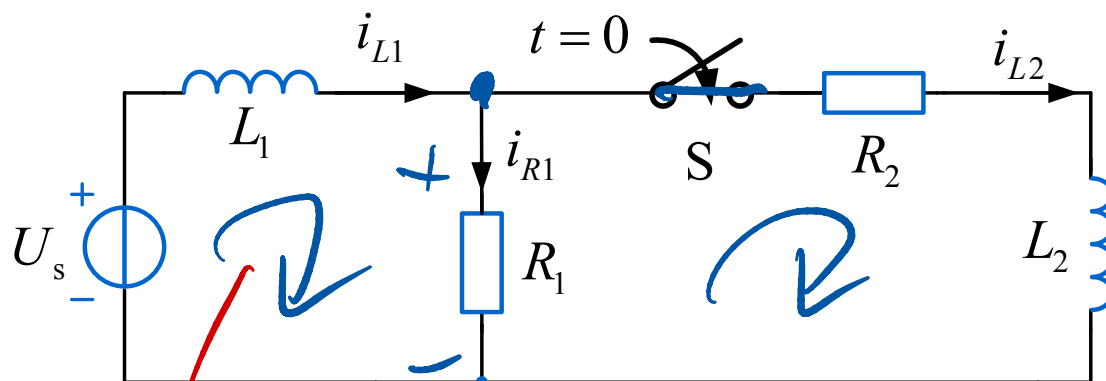
电路原已达稳态，
求开关闭合后的
两个电感电流。



8 二阶动态电路

例题1 (提高)

$$U_s = 6\text{V}, R_1 = 2\Omega, R_2 = 6\Omega, L_1 = 1\text{H}, L_2 = 4\text{H}$$



电路原已达稳态，
求开关闭合后的
两个电感电流。

$$-R_1 i_{R1} + R_2 i_{L2} + L_2 \frac{di_{L2}}{dt} = 0 \quad \text{无法求出 } \frac{di_{L2}}{dt}$$

$$-U_s + L_1 \left(\frac{di_{L1}}{dt} \right) + R_1 i_{R1} = 0 \quad i_{R1} = i_{L1} - i_{L2}$$

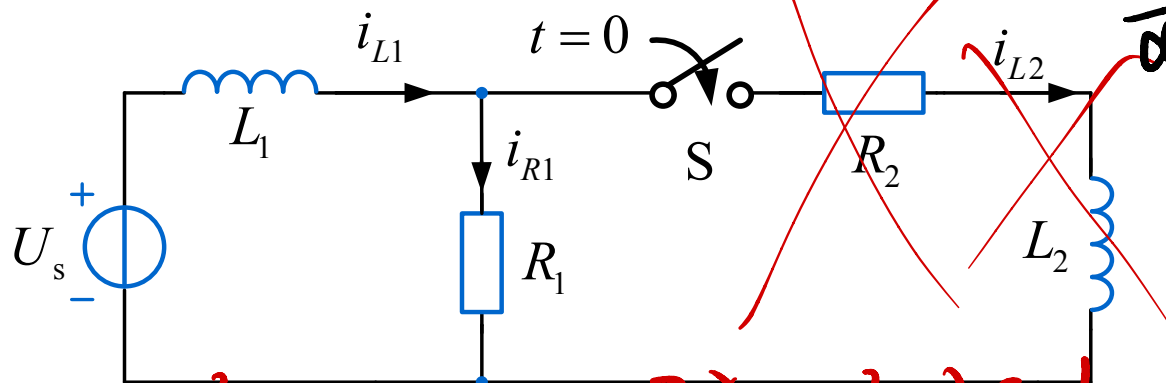
$$-U_s + L_1 \left[\frac{di_{L1}}{dt} \right] + R_1 (i_{L1} - i_{L2}) = 0$$

$$\frac{di_{L1}}{dt} = \frac{U_s}{L_1} + \frac{R_1}{L_1} (i_{L1} - i_{L2}) \Rightarrow \frac{di_{L1}}{dt} \Big|_{t=0+} = \frac{6}{1} + \frac{2}{1} (1 - 0) = 8$$

8 二阶动态电路

例题1 (提高)

$$U_s = 6\text{V}, R_1 = 2\Omega, R_2 = 6\Omega, L_1 = 1\text{H}, L_2 = 4\text{H}$$



$$0 = -3A - B \quad (2)$$

$$\frac{di_L}{dt} = A(2)e^{-3t} + B(-1)e^{-t}$$

电路原已达稳态，
求开关闭合后的
两个电感电流。

$$3 = 4 + A + B \quad (1)$$

$$x^2 + 4\lambda + 3 = 0 \Rightarrow \lambda_1 = -3 \quad \lambda_2 = -1$$

$$\frac{d^2 i_{L1}}{dt^2} + 4 \frac{di_{L1}}{dt} + 3i_{L1} = 12$$

$$i_{L1}(t) = i_{L1}^{(1)} + i_{L1}^{(2)} = 4 + (Ae^{-3t} + Be^{-t}) \text{ A}$$

$$i_{L1}(0_+) = i_{L1}(0_-) = \left(\frac{U_s}{R_1}\right) = \frac{6}{2} = 3 \text{ A}$$

$$i_{L1}(0_+) = i_{L1}(0_-) \quad i_{L2}(0_+) = i_{L2}(0_-) = 0 \text{ A}$$

$$-U_s + L_1 \frac{di_{L1}}{dt} + R_1(i_{L1} - i_{L2}) = 0$$

$$-6 + 1 \times \left. \frac{di_{L1}}{dt} \right|_{t=0_+} + 2 \times (3 - 0) = 0$$

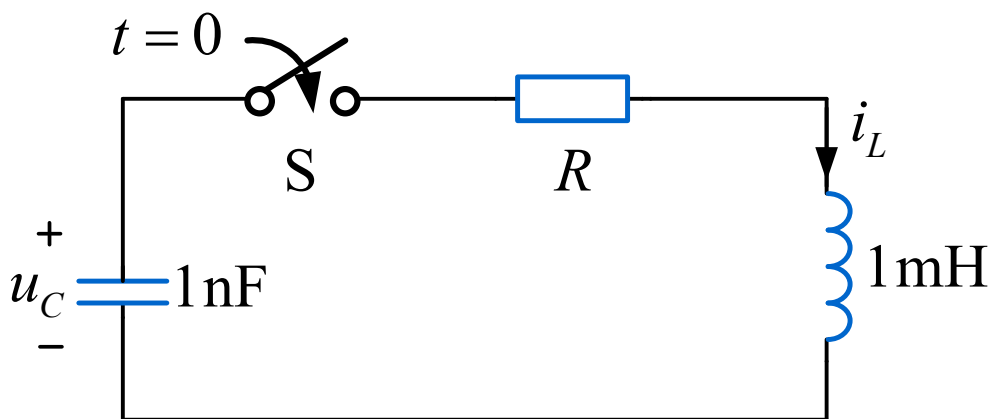
$$i_{L1}(t) = 4 + 0.5e^{-3t} - 1.5e^{-t} \text{ A}$$

$$i_{L2}(t) = 1 - 0.25e^{-3t} - 0.75e^{-t} \text{ A}$$

$$\left. \frac{di_{L1}}{dt} \right|_{t=0_+} = 0$$

$$3 = 4 + A + B$$

同步练习题1 (提高)

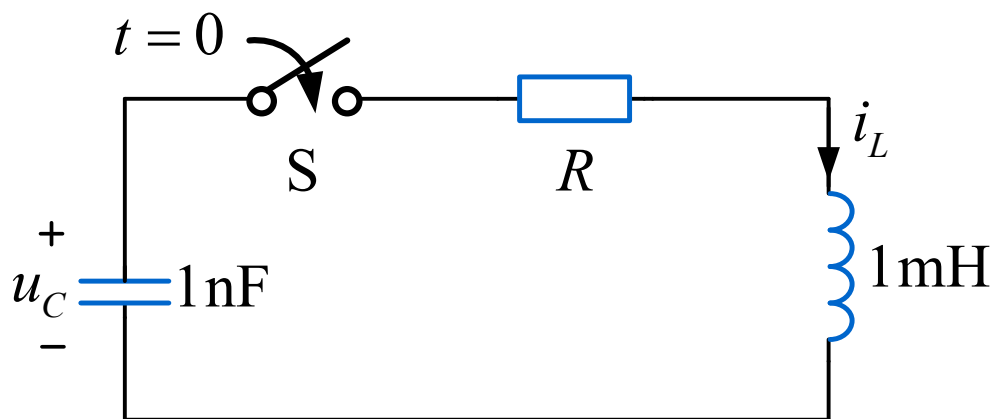


电路原已达稳态， $u_C(0_-)=2\text{ V}$ 。

开关闭合后， R 为何值时二阶电路工作于临界阻尼状态？求此时的电容电压和电感电流。



同步练习题1 (提高)



电路原已达稳态， $u_C(0_-) = 2 \text{ V}$ 。

开关闭合后， R 为何值时二阶电路工作于临界阻尼状态？求此时的电容电压和电感电流。

答案： $R = 2 \text{ k}\Omega$, $u_C(t) = (2 + 2 \times 10^6 t) e^{-10^6 t} \text{ V}$, $i_L(t) = 2 \times 10^3 t e^{-10^6 t} \text{ A}$

- 二阶电路的微分方程是二阶微分方程
- 二阶微分方程的求解有**固定的步骤**
- 二阶微分方程**解的形式**由特征方程的特征根决定
- **代入初始条件**，确定待定系数时二阶微分方程求解的**难点**！
- 零输入RLC串联二阶电路可能的工作状态有4种：
过阻尼（衰减）；临界阻尼；欠阻尼（振荡衰减）；零阻尼（振荡）
- 二阶电路的求解过程很长，需要**耐心和细心**



感谢大家聆听

主讲人：邹建龙

时 间：2025年4月10日

